

Abdullah Arafat Miah

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EDUCATION

University of Rhode Island

Ph.D. in Electrical Engineering | GPA: 3.96/4.00

Kingston, RI, USA

09/2023 – 05/2028 (Expected)

Dissertation: *Investigating Backdoor Attacks and Defense Frameworks for the Secure Deployment of Modern Neural Networks.*

RESEARCH EXPERIENCE

Graduate Research Assistant

University of Rhode Island

Kingston, RI, USA

Sep 2023 – Present

Bad-VL-RAG: Backdoor Attacks on Vision-Language RAG | VLMs, RAG, PyTorch, FAISS, Python

- Built an end-to-end vision-language retrieval-augmented generation (RAG) pipeline to enhance VLMs on knowledge-intensive tasks, then studied its robustness by proposing novel backdoor attacks that exploit the knowledge-poisoning vulnerabilities of RAG systems through white-box and black-box trigger optimization.

Backdoor Defense: Pruning- & Purification-based Mitigation | GANs, Diffusion Models, ViTs, CNNs, PyTorch, Python

- Built a network-pruning defense that ranks and prunes backdoor neurons via adversarial proxies, supported by a neuron-distinguishability theoretical framework.
- Built a black-box backdoor-defense framework using a two-stage input-purification method, running 34–123× faster than prior methods.

SNN Backdoor Attacks: T-Backdoor & BadSNN | SNNs, Neuromorphic/Event Data, SpikingJelly, snnTorch, Python

- Proposed spike-preserving backdoors that exploit temporal redundancy (T-Backdoor) and LIF neuron dynamics (BadSNN) to attack SNNs on static-image and neuromorphic vision datasets.

Architectural Backdoors on LLMs & Perplexity-based Defenses | LLMs, PyTorch, Python

- Proposed a training-free backdoor attack on LLMs by embedding an attack module within the model's architecture.
- Implemented a perplexity-based defense that detects trigger tokens in LLM backdoors; showed defense efficacy degrades sharply as trigger length grows.

SELECTED PUBLICATIONS

- Abdullah Arafat Miah**, Yu Bi. “Bad-VL-RAG: Backdoor Attacks against Vision-Language Retrieval-Augmented Generation.” *In preparation for submission to CVPR 2027*.
- Abdullah Arafat Miah**, Yu Bi. “DeepCurer: Pruning-based Backdoor Mitigation via Progressive Neuron Ranking using Adversarial Proxies.” *Neurocomputing*, 2026, art. 134409.
- Abdullah Arafat Miah**, Yu Bi. “Lite-BD: A Lightweight Black-box Backdoor Defense via Reviving Multi-Stage Image Transformations.” *IEEE/INNS IJCNN 2026* (Accepted).
- Abdullah Arafat Miah**, Kevin Vu, Yu Bi. “T-Backdoor: Exploiting Temporal Redundancy in Neuromorphic Data for Spike-preserving Backdoor Attacks on SNNs.” *IEEE Transactions on Artificial Intelligence* (Under Review).
- Abdullah Arafat Miah**, Yu Bi. “NoiseAttack: An Evasive Sample-Specific Multi-Targeted Backdoor Attack Through White Gaussian Noise.” *International Journal of Information Technology* (Accepted).
- Abdullah Arafat Miah**, Yu Bi. “BadSNN: Backdoor Attacks on Spiking Neural Networks via Adversarial Spiking Neuron.” *IEEE Transactions on Multimedia* (Under Review).
- Md Nabi Newaz Khan, **Abdullah Arafat Miah**, Yu Bi. “BadImplant: Injection-based Multi-Targeted Graph Backdoor Attack.” *Journal of Information Security and Applications* (Under Review).
- Abdullah Arafat Miah**, Yu Bi. “TrojanNLP: Exploiting the Vulnerability of Natural Language Models via Defense-Aware Architectural Backdoor.” *SN Computer Science* (Under Review).

TECHNICAL SKILLS

- Languages & Tools:** Python, C++, C, R, Julia, Bash, PyTorch, TensorFlow, Hugging Face (Transformers/Diffusers), Scikit-Learn, OpenCV, FAISS, SpikingJelly, snnTorch, Git, Linux, HPC/GPU, MATLAB
- Models:** VLMs, LLMs, GANs, Diffusion Models, ViTs, SNNs, CNNs, LSTMs
- Research Areas:** Adversarial ML & AI Security, Multimodal RAG, AI Agents, Computer Vision, Privacy-preserving ML

TEACHING EXPERIENCE

Graduate Teaching Assistant

University of Rhode Island — Dept. of Electrical, Computer & Biomedical Engineering

Kingston, RI, USA

Sep 2024 – Present

- Lead undergraduate lab sections for Digital Circuit Design (ELE 202) and Digital Circuit Design with FPGAs (ELE 302) — lab preparation, instruction, grading, and exam proctoring.

PROFESSIONAL SERVICE

- Reviewer**, IEEE/INNS International Joint Conference on Neural Networks (IJCNN), 2026.